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Inventor(s)

FIDALEO; Douglas A.

ENSURING USER DATA SECURITY WHILE PERSONALIZING A SOCIAL AGENT

Abstract

A social agent system includes a computing platform having processing hardware and a system memory storing a social agent software code. The processing hardware is configured to execute the social agent software code to receive, from a client system, input data provided by a user of the client system when the user is interacting with the social agent system using the client system, and to select, using the input data, a dialogue template for responding to the user. The dialogue template includes one or more placeholder fields to be filled by the client system to create a personalized dialogue for responding to the user. The processing hardware is further configured to execute the social agent software code to deliver, to the client system, the dialogue template including the one or more placeholder fields to be filled by the client system to create the personalized dialogue for responding to the user.

Inventors: FIDALEO; Douglas A. (Santa Clarita, CA)

Applicant: DISNEY ENTERPRISES, INC. (Burbank, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of the co-pending U.S. patent application titled, “Ensuring User Data Security While Personalizing a Social Agent”, filed on Jun. 11, 2021, and having Ser. No. 17/345,671. The subject matter of this related application is hereby incorporated herein by reference.

BACKGROUND

[0002] A characteristic feature of human social interaction is personalization. For example, when two people interact repeatedly in a similar manner, such as greeting one another, a variety of different expressions may be used despite the fact that a generic “hello” would be adequate in almost every instance. Instead, a human speaker may select expressions for use in an interaction with another person based on whether that person is a child, a teenager, or an adult, as well as based on their personal history of interactions with the other person. In order for a non-human social agent to engage in a realistic interaction with a user, it is desirable that the social agent also be capable of varying its form of expression in a personalized and seemingly natural way.

[0003] Personalizing the responsiveness of a social agent during an interaction with a particular user has typically required the system providing the social agent to acquire and store user data as part of an overall user profile including demographic and geographic details describing the user, as well as a history of previous interactions with the user. However, concerns regarding personal privacy, data security, and liability for data breaches make it increasingly advantageous, for users and system administrators alike, to provide users with the ability to restrict access to their personal user data while continuing to enable use of such data to personalize their interactions with a social agent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 shows a diagram of a social agent system, according to one exemplary implementation;

[0005] FIG. 2 shows a diagram including a more detailed exemplary representation of a user system in interaction with the social agent system shown in FIG. 1, according to one implementation;

[0006] FIG. 3A shows a more detailed diagram of an input unit suitable for use in an interaction with the social agent system shown in FIGS. 1 and 3, according to one implementation;

[0007] FIG. 3B shows a more detailed diagram of an output unit suitable for use in an interaction with the social agent system shown in FIGS. 1 and 3, according to one implementation;

[0008] FIG. 4 shows a flowchart presenting an exemplary method for use by a social agent system to interact with a user, according to one implementation; and

[0009] FIG. 5 shows a flowchart presenting an exemplary method for execution by a social agent client application, according to one implementation.

DETAILED DESCRIPTION

[0010] The following description contains specific information pertaining to implementations in the present disclosure. One skilled in the art will recognize that the present disclosure may be

implemented in a manner different from that specifically discussed herein. The drawings in the present application and their accompanying detailed description are directed to merely exemplary implementations. Unless noted otherwise, like or corresponding elements among the figures may be indicated by like or corresponding reference numerals.

[0011] As stated above, a characteristic feature of human social interaction is personalization. For example, and as also stated above, when two people interact repeatedly in a similar manner, such as greeting one another, a variety of different expressions may be used despite the fact that a generic “hello” would be adequate in almost every instance. Instead, a human speaker may select expressions for use in an interaction with another person based on whether that person is a child, a teenager, or an adult, as well as based on their personal history of interactions with the other person. In order for a non-human social agent to engage in a realistic interaction with a user, it is desirable that the social agent also be capable of varying its form of expression in a personalized and seemingly natural way. However, due to concerns regarding personal privacy, data security, and liability for data breaches, there is a need in the art for a data management solution that provides users with the ability to restrict access to their personal user data while continuing to enable use of such data to personalize their interactions with a social agent.

[0012] The present application is directed to automated systems and methods that address and overcome the deficiencies in the conventional art described above. The inventive concepts disclosed in the present application advantageously enable the automated determination of naturalistic expressions for use by a social agent in responding to an interaction with a user, while concurrently enabling the user to selectively sequester their personal user data from the system providing the social agent. It is noted that, as defined in the present application, the term “response” may refer to language based expressions, such as a statement or question, or to non-verbal expressions. Moreover, the term “non-verbal expression” may refer to vocalizations that are not language based, i.e., non-verbal vocalizations, as well as to physical gestures and postures. Examples of non-verbal vocalizations may include a sigh, a murmur of agreement or disagreement, or a giggle, to name a few.

[0013] As defined in the present application, the term “goal” refers to an aim or objective motivating a user to interact with a social agent. Examples of goals include the acquisition of information, arrival at a particular destination, or viewing or participating in an event, to name a few. Furthermore, because it is not possible to have definite knowledge of a human user's inner mental state, as used in the present application the term goal is to be interpreted as “inferred goal.” Thus, as used herein, the “goal of the user” refers to the “inferred goal of the user.”

[0014] Furthermore, as defined in the present application, the terms “automation,” “automated,” and “automating” refer to systems and processes that do not require human intervention. Although in some implementations a human editor may review the personalized responses generated by the systems and using the methods described herein, that human involvement is optional. Thus, the methods described in the present application may be performed under the control of hardware processing components of the disclosed automated systems.

[0015] In addition, as defined in the present application, the term “social agent” refers to a non-human communicative entity rendered in hardware and software that is designed for expressive interaction with a human user. In some use cases, a social agent may take the form of a virtual character rendered on a display (i.e., social agent **116a** rendered on display **158** of client system **150**, in FIG. **1**) and appearing to watch and listen to a user in order to respond to a communicative user input. In other use cases, a social agent may take the form of a machine (i.e., social agent **116b**, in FIG. **1**), such as a robot for example, appearing to watch and listen to the user in order to respond to a communicative user input. Alternatively, a social agent may be implemented as an automated voice response (AVR) system, or an interactive voice response (IVR) system, for example.

[0016] FIG. **1** shows a diagram of a social agent system, according to one exemplary

implementation. As shown in FIG. 1, social agent system **100** (hereinafter “system **100**”) includes computing platform **102** having processing hardware **104** and system memory **106** implemented as a non-transitory storage medium. According to the present exemplary implementation, system memory **106** stores social agent software code **110** and dialogue template database **120** storing dialogue templates **122a**, **122b**, and **122c** (hereinafter “dialogue templates **122a-122c**”). In addition, FIG. 1 shows social agents **116a** and **116b** shown respectively as a virtual character and a physical machine such as a robot.

[0017] As further shown in FIG. 1, system **100** is implemented within a use environment including communication network **112** providing network communication links **114**, as well as client system **150** including display **158** and utilized by user **118** to interact with system **100**. Also shown in FIG. 1 are input data **128** corresponding to an interaction by user **118** with social agent **116a** or **116b**, selected dialogue template **122** from among dialogue templates **122a-122c** delivered to client system **150** by system **100**, personalized dialogue **148**, created by client system **150** using selected dialogue template **122**, and optional wireless communication link **115** between client system **150** and social agent **116b**.

[0018] Although the present application may refer to one or both of social agent software code **110** and dialogue template database **120** as being stored in system memory **106** for conceptual clarity, more generally, system memory **106** may take the form of any computer-readable non-transitory storage medium. The expression “computer-readable non-transitory storage medium,” as defined in the present application, refers to any medium, excluding a carrier wave or other transitory signal that provides instructions to processing hardware **104** of computing platform **102**. Thus, a computer-readable non-transitory medium may correspond to various types of media, such as volatile media and non-volatile media, for example. Volatile media may include dynamic memory, such as dynamic random access memory (dynamic RAM), while non-volatile memory may include optical, magnetic, or electrostatic storage devices. Common forms of computer-readable non-transitory storage media include, for example, optical discs, RAM, programmable read-only memory (PROM), erasable PROM (EPROM), and FLASH memory.

[0019] It is further noted that although FIG. 1 depicts social agent software code **110** and dialogue template database **120** as being co-located in system memory **106**, that representation is also merely provided as an aid to conceptual clarity. More generally, system **100** may include one or more computing platforms **102**, such as computer servers for example, which may be co-located, or may form an interactively linked but distributed system, such as a cloud-based system, for instance. As a result, processing hardware **104** and system memory **106** may correspond to distributed processor and memory resources within system **100**.

[0020] Processing hardware **104** may include multiple hardware processing units, such as one or more central processing units, one or more graphics processing units, one or more tensor processing units, one or more field-programmable gate arrays (FPGAs), custom hardware for machine-learning training or inferencing, and an application programming interface (API) server, for example. By way of definition, as used in the present application, the terms “central processing unit” (CPU), “graphics processing unit” (GPU), and “tensor processing unit” (TPU) have their customary meaning in the art. That is to say, a CPU includes an Arithmetic Logic Unit (ALU) for carrying out the arithmetic and logical operations of computing platform **102**, as well as a Control Unit (CU) for retrieving programs, such as software code **110**, from system memory **106**, while a GPU may be implemented to reduce the processing overhead of the CPU by performing computationally intensive graphics or other processing tasks. A TPU is an application-specific integrated circuit (ASIC) configured specifically for artificial intelligence (AI) applications such as machine learning modeling.

[0021] It is noted that, as defined in the present application, the expression “machine learning model” may refer to a mathematical model for making future predictions based on patterns learned from samples of data or “training data.” Various learning algorithms can be used to map

correlations between input data and output data. These correlations form the mathematical model that can be used to make future predictions on new input data. Such a predictive model may include one or more logistic regression models, Bayesian models, or neural networks (NNs). Moreover, a “deep neural network,” in the context of deep learning, may refer to an NN that utilizes multiple hidden layers between input and output layers, which may allow for learning based on features not explicitly defined in raw data.

[0022] In some implementations, computing platform **102** may correspond to one or more web servers, accessible over a packet-switched network such as the Internet, for example. Alternatively, computing platform **102** may correspond to one or more computer servers supporting a private wide area network (WAN), local area network (LAN), or included in another type of limited distribution or private network. Moreover, in some implementations, communication network **112** may be a high-speed network suitable for high performance computing (HPC), for example a **10** Gige network or an Infiniband network.

[0023] Although client system **150** is shown as a smartphone in FIG. **1**, that representation is provided merely as an example as well. More generally, client system **150** may be any suitable mobile or stationary computing device or system that includes display **158**, provides a user interface, supports communications via communication network **112** and wireless communication link **115**, and implements data processing capabilities sufficient to implement the functionality ascribed to client system **150** herein. For example, in other implementations, client system **150** may take the form of a tablet computer, laptop computer, desktop computer, smart television (smart TV), game console, or a smart wearable device, such as a smartwatch, for instance. With respect to display **158** of client system **150**, display **158** may be implemented as a liquid crystal display (LCD), light-emitting diode (LED) display, organic light-emitting diode (OLED) display, quantum dot (QD) display, or any other suitable display screen that perform a physical transformation of signals to light.

[0024] By way of overview of the functionality of social agent system **100** and client system **150**, in a typical use case, social agent system **100** may receive, from client system **150**, input data **128** provided by user **118** of the client system **150** when user **118** is interacting with social agent system **100** using client system **150**. Processing hardware **104** of computing platform **102** may execute social agent software code **110** to select, using input data **128**, one of dialogue templates **122a-122b** (i.e., selected dialogue template **122**) for responding to user **118**. Selected dialogue template **122**, as well as dialogue templates **122a-122c**, each includes one or more placeholder fields to be filled by client system **150** using user data stored on client system **150** and inaccessible to social agent system **100**, to create a personalized dialogue for responding to user **118**. Social agent system **100** may then deliver, to client system **150**, selected dialogue template **122** including the one or more placeholder fields to be filled by client system **150** to create personalized dialogue **148**, using selected dialogue template **122**.

[0025] Client system **150** may receive selected dialogue template **122** from social agent system **100**, and may identify user data stored locally on client system **150** for filling the one or more placeholder fields included in selected dialogue template **122**. Client system may then fill the one or more placeholder fields using the user data to create personalized dialogue **148** for responding to user **118**, and may execute the personalized dialogue, using one or more output devices of client system **150**, such as display **158**, or using wireless communication link **115** to control output devices of social agent **116b**.

[0026] FIG. **2** shows a more detailed representation of exemplary client system **250** in combination with computing platform **202** of social agent system **200**, according to one implementation. As shown in FIG. **2**, client system **250** is communicatively coupled to computing platform **202** by network communication link **214**. Computing platform **202** includes processing hardware **204**, system memory **206** implemented as a non-transitory storage medium, and transceiver **208**. According to the exemplary implementation depicted in FIG. **2**, system memory **206** stores social

agent software code **210** and dialogue template database **220** including exemplary dialogue templates **222a**, **222b**, and **222c** (hereinafter “dialogue templates **222a-222c**”).

[0027] Client system **250** includes hardware processor **254** and memory **256** implemented as a non-transitory storage medium storing user database **264** and social agent client application **266**. As also shown in FIG. 2, client system **250** may include any or all of transceiver **252**, input unit **230**, and output unit **240** receiving personalized dialogue **248** from social agent client application **266**. Also shown in FIG. 2 are input data **228** generated by client system **250** and received by social agent system **200** via network communication link **214**, and selected dialogue template **222** delivered to client device **250** by social agent system **200**, also via network communication link **214**.

[0028] Social agent system **200** including computing platform **202** having processing hardware **204** and system memory **206** storing dialogue template database **220** including dialogue templates **222a-222c** corresponds in general to social agent system **100** including computing platform **102** having processing hardware **104** and system memory **106** storing dialogue template database **120** including dialogue templates **122a-122c**, in FIG. 1. Thus, social agent system **200** may share any of the characteristics attributed to social agent system **100** by the present disclosure, and vice versa. That is to say, although not shown in FIG. 1, like social agent system **200**, social agent system **100** may include transceiver **208**.

[0029] In addition, social agent software code **210** corresponds in general to social agent software code **110**, in FIG. 1. Consequently, social agent software code **110** may share any of the characteristics attributed to social agent software code **210** by the present disclosure, and vice versa. Moreover, input data **228**, selected dialogue template **222**, personalized dialogue **248**, and network communication link **214**, in FIG. 2, correspond respectively in general to input data **128**, selected dialogue template **122**, personalized dialogue **148**, and network communication links **114**, in FIG. 1, and those corresponding features may share any of the characteristics attributed to either corresponding feature herein.

[0030] Client system **250** corresponds in general to client system **150**, and those corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure. Thus, like client system **250**, client system **150** may include features corresponding to hardware processor **254**, input unit **230**, output unit **240**, transceiver **252**, and memory **256** storing user database **264** and social agent client application **266**. It is noted that display **158** of client system **150** may be included as a component of one or both of input unit **230** and output unit **240**.

[0031] Transceiver **208** and transceiver **252** may be implemented as wireless communication hardware and software enabling computing platform **102/202** of social agent system **100/200** to exchange data with client system **150/250** via network communication link **114/214**. For example, transceiver **208** and transceiver **252** may be implemented as fourth generation of broadband cellular technology (4G) wireless transceivers, or as 5G wireless transceivers configured to satisfy the IMT-2020 requirements established by the International Telecommunication Union (ITU). Alternatively, or in addition, transceiver **208** and transceiver **252** may be configured to communicate via one or more of WiFi, Bluetooth, ZigBee, and 60 GHz wireless communications methods.

[0032] With respect to social agent client application **266**, it is noted that social agent client application **266** may be a client application of social agent software code **110/210**. It is noted that, in some implementations, one or both of social agent software code **110/210** and social agent client application **266** may include one or more machine learning models, such as NNs, for example. Social agent client application **266** is configured to create personalized dialogue **148/248** for user **118** by filling one or more placeholder fields included in selected dialogue template **122/222**, selected by social agent software code **110/210**, utilizing user data obtained from user database **264**. According to various implementations of the present disclosure, social agent client application **266** enables user **118** to selectively restrict access to the user data stored locally on client system **150/250** by social agent system **100/200**, while advantageously enabling use of that restricted user

data to create personalized dialogue **148/248** from selected dialogue template **122/222**.

[0033] According to the exemplary implementation shown in FIG. 2, social agent client application **266** is located in memory **256** of client system **150/250**, subsequent to transfer of social agent client application **266** to client system **150/250** over a packet-switched network, such as the Internet, for example. Once present on client system **150/250**, social agent client application **266** may be persistently stored in memory **256** and may be executed locally on client system **150/250** by hardware processor **254**.

[0034] FIG. 3A shows a more detailed diagram of input unit **330**, according to one implementation. As shown in FIG. 3A, input unit **330** includes input device **332** such as a keyboard, trackpad, or mouse, for example, sensors **334**, one or more microphones **335** (hereinafter “microphone(s) **335**”), analog-to-digital converter (ADC) **336**, and may include display **358** in the form of a touchscreen. As further shown in FIG. 3A, sensors **334** of input unit **330** may include radio-frequency identification (RFID) sensor **334a**, facial recognition (FR) sensor **334b**, automatic speech recognition (ASR) sensor **334c**, object recognition (OR) sensor **334d**, and one or more cameras **334e** (hereinafter “camera(s) **334e**”). Input unit **330** corresponds in general to input unit **230**, in FIG. 2. Thus, input unit **230** and may share any of the characteristics attributed to input unit **330** by the present disclosure, and vice versa.

[0035] It is noted that the specific sensors shown to be included among sensors **334** of input unit **230/330** are merely exemplary, and in other implementations, sensors **334** of input unit **230/330** may include more, or fewer, sensors than RFID sensor **334a**, FR sensor **334b**, ASR sensor **334c**, OR sensor **334d**, and camera(s) **334e**. Moreover, in other implementations, sensors **334** may include a sensor or sensors other than one or more of RFID sensor **334a**, FR sensor **334b**, ASR sensor **334c**, OR sensor **334d**, and camera(s) **334e**. It is further noted that camera(s) **334e** may include various types of cameras, such as red-green-blue (RGB) still image and video cameras, RGB-D cameras including a depth sensor, and infrared (IR) cameras, for example.

[0036] FIG. 3B shows a more detailed diagram of output unit **340**, according to one implementation. As shown in FIG. 3B, output unit **340** includes display **358**, Text-To-Speech (TTS) module **342** and one or more audio speakers **344** (hereinafter “audio speaker(s) **344**”). As further shown in FIG. 3B, in some implementations, output unit **340** may include one or more mechanical actuators **346** (hereinafter “mechanical actuator(s) **346**”), which may include one or more haptic actuators **347** (hereinafter “haptic actuator(s) **347**”). It is noted that, when included as a component or components of output unit **340**, mechanical actuator(s) **346** may be used to produce facial expressions by social agent **116b**, and to articulate one or more limbs or joints of social agent **116b**.

[0037] Output unit **340** corresponds in general to output unit **240**, in FIG. 2. Thus, output unit **240** may share any of the characteristics attributed to output unit **340** by the present disclosure, and vice versa. Moreover, display **358** shown in FIGS. 3A and 3B corresponds in general to display **158**, in FIG. 1. Thus, like display **158**, display **358** may take the form of an LCD, LED display, OLED display, QD display, or any other suitable display screen that perform a physical transformation of signals to light.

[0038] It is noted that the specific components shown to be included in output unit **240/340** are merely exemplary, and in other implementations, output unit **240/340** may include more, or fewer, components than display **208**, TTS module **342**, audio speaker(s) **344**, and mechanical actuator(s) **346**. Moreover, in other implementations, output unit **240/340** may include a component or components other than one or more of display **208**, TTS module **342**, audio speaker(s) **344**, and mechanical actuator(s) **346**. It is further noted that in implementations in which social agent **116b** takes the form of a robot or other type of physical machine, an output unit corresponding to output unit **240/340** may be integrated with social agent **116b**, as well as with client system **150/250**. In other words, in some implementations, social agent **116b** may include output unit **240/340**.

[0039] The functionality of social agent software code **110/210** and social agent client application **266** is further described below by reference to FIGS. 4 and 5. FIG. 4 shows flowchart **470**

presenting an exemplary method for use by social agent system **100/200** to interact with user **118**, according to one implementation, while FIG. 5 shows flowchart **580** presenting an exemplary method for execution by a social agent client application, according to one implementation. With respect to the actions outlined in FIGS. 4 and 5, it is noted that certain details and features have been left out of respective flowchart **470** and flowchart **580** in order not to obscure the discussion of the inventive features in the present application.

[0040] Referring to FIG. 4 in combination with FIGS. 1 and 2, flowchart **470** begins with receiving, from client system **150/250**, input data **128/228** provided by user **118** of client system **150/250** when user **118** is interacting with social agent system **100/200** using client system **150/250** (action **471**). Input data **128/228** may be received by social agent software code **110/210**, executed by processing hardware **104/204** of computing platform **102/202**, via communication network **112** and network communication links **114/214**. Input data **128/228** may be received in the form of speech or text by user **118** in interacting with social agent **116a** or **116b**, for example.

[0041] Alternatively, or in addition, input data **128/228** may be received in the form of one or more non-verbal expressions by user **118** in interacting with social agent **116a** or **116b**. As noted above, the term non-verbal expression may refer to vocalizations that are not language based, i.e., non-verbal vocalizations, as well as to physical gestures and physical postures. Examples of non-verbal vocalizations may include a sigh, a murmur of agreement or disagreement, or a giggle, to name a few. Alternatively, input data **128/228** may be received as speech uttered by user **118**, or as one or more manual inputs to input device **332** in the form of a keyboard or touchscreen, for example, by user **118**. Thus, the interaction with social agent system **110/200** by user **118** using client system **150/250** may include one or more of text input by user **118**, speech by user **118**, a non-verbal vocalization by user **118**, a facial expression by user **118**, a gesture by user **118**, or a physical posture of user **118**.

[0042] According to various implementations, client system **150/250** advantageously includes input unit **230/330**, which may obtain video and perform motion capture, using camera(s) **334e** for example, in addition to capturing audio using microphone(s) **335**. As a result, social agent client application **266**, when executed by hardware processor **254** of client system **150/250**, may receive audio, video, and motion capture features from input unit **230/330**, and may detect a variety of verbal and non-verbal expressions by user **118** in an with social agent system **100/200** using client system **150/250**. Input data **128/228** may then be conveyed to social agent system **100/200**.

[0043] In some implementations, as shown in FIG. 4, flowchart **470** may further include inferring, in response to receiving input data **128/228**, a goal of user **118** (action **472**). For example, based on a verbal expression, a non-verbal expression, or a combination of verbal and non-verbal expressions described by input data **128/228**, processing hardware **104/204** may execute social agent software code **110/220** to determine a goal of user **118**, for example, using a machine learning predictive model included in social agent software code **110/210**. The goal of user **118** may be determined based on the subject matter of the interaction described by input data **128/228**. As noted above, the term goal refers to the aim or objective motivating user **118** to interact with social agent system **100/200**, and may include the acquisition of information, arrival at a particular destination, or viewing or participating in an event, to name a few examples.

[0044] Flowchart **470** further includes selecting, using at least input data **128/228**, one of dialogue templates **122a-122c/222a-222c** for responding to user **118**, the selected dialogue template **122/222** including at least one placeholder field configured to be filled by client system **150/250** to create personalized dialogue **148/248**, using the selected dialogue template **122/222**, for responding to user **118** (action **473**). Thus, selected dialogue template **122/222** is selected in action **473** from among multiple predetermined dialogue templates **122a-122c/222a-222c**. Action **473** may be performed by social agent software code **110/210**, executed by processing hardware **104/204** of computing platform **102/202**.

[0045] It is noted that, in some implementations, the method outlined by flowchart **470** may omit

action **472**, and action **473** may follow directly from action **471**. In those implementations, action **473** may be performed using input data **128/228** alone. However, in implementations in which action **472** is performed, action **473** may include using the inferred goal of user **118**, as well as input data **128/228**, to select the dialogue template for use in responding to user **118**.

[0046] Flowchart **470** further includes delivering, to client system **150/250**, selected dialogue template **122/222** including the at least one placeholder field to be filled by client system **150/250** to create personalized dialogue **148/248**, using selected dialogue template **122/222**, for responding to user **118** (action **474**). Action **474** may be performed by social agent software code **110/210**, executed by processing hardware **104/204** of computing platform **102/202**, as a result of communication with client system **150/250** via communication network **112** and network communication links **114/214**.

[0047] By way of example, dialogue template **122/222** selected in action **473** may include the following greeting: “Hi there (name placeholder field). How are things going in (location placeholder field)?” The user data for filling the name and location placeholder fields in this exemplary dialogue template may not be available to social agent system **100/200** due to that user data being selectably rendered inaccessible to social agent system **100/200** by user **118** through use of social agent client application **266**. As a result, that user data may be sequestered on client system **150/250** and may be used locally on client system **150/250** by social agent client application **266** in action **474** to create personalized dialogue **148/248**, using selected dialogue template **122/222**, for responding to user **118**. For instance, where user **118** is named Mary Clark and Mary is located in Santa Clarita, California, personalized dialogue **148/248** based on selected dialogue template **122/222** may state: “Hi there (Mary). How are things going in (Santa Clarita)?”

[0048] In some implementations, the privacy level restricting access to the user data stored in user database **264** of client system **150/250** may be predetermined and fixed. For example, in some of those implementations, the user data stored in user database **264** may include personally identifiable information (PII) of user **118**, and social agent client application **266** may be configured to make all PII unavailable to social agent system **100/200**. However, in other implementations, it may be advantageous or desirable to enable user **118** to control their preferred level of privacy by selecting that privacy level from multiple predetermined alternative privacy levels using social agent client application **266**. For example, in one implementation, one user controllable privacy level may make all user data stored in user database **264** available to social agent system **100/200**, while another user controllable privacy level may not make any user data available to social agent system **100/200**. One or more user selectable intermediate privacy levels may partially anonymize the user data made available to social agent system **100/200**. By way of example, where user **118** is twenty-five year old Mary Clark in Santa Clarita, Mary may control her privacy level using social agent client application **266** by selecting an intermediate privacy level allowing social agent system **100/200** to obtain user data describing Mary as an adult female user in California, or as a female user in the United States.

[0049] In some implementations, social agent client application **266** may be configured to partially anonymize user data other than PII, using a proto-object as a proxy for a famous or otherwise identifiable object favored by user **118**. That is to say, the proto-object corresponds to a preference of user **118**. For example, where a favorite character of user **118** is a world famous animated mouse, social agent client application **266** may make user data describing user **118** as having a favorite mouse character available to social agent system **100/200** without identifying the famous animated mouse per se.

[0050] Moving to FIG. 5, and Referring to flowchart **580** in combination with FIGS. 1 and 2, flowchart **580** begins with sending, to social agent system **100/200** including social agent software code **110/210**, input data **128/228** provided by user **118** of client system **150/250** when user **118** is interacting with social agent system **1100/200** using client system **150/250** (action **581**). Input data **128/228** may be sent to social agent system **100/200** by social agent client application **266**,

executed by hardware processor **254** of client system **150/250**, via communication network **112** and network communication links **114/214**. Input data **128/228** may be received in the form of speech or text by user **118** in interacting with social agent **116a** or **116b**, for example. Input data **128/228** may be collected and conveyed in the manner discussed above by reference to action **471** of flowchart **470**.

[0051] Flowchart **580** further includes receiving, from social agent system **100/200**, dialogue template **122/222** for responding to user **118**, dialogue template **122/222** including at least one placeholder field configured to be filled by client system **150/250** (action **582**). As noted above, dialogue template **122/222** is selected in action **473** described above by social agent software code **110/210** of social agent system **100/200** from among multiple predetermined dialogue templates **122a-122c/222a-222c**. Selected dialogue template **122/222** may be received in action **582** by social agent client application **266**, executed by hardware processor **254** of client system **150/250**, via communication network **112** and network communication links **114/214**.

[0052] Flowchart **580** further includes identifying user data stored on client system **150/250** for filling the one or more placeholder fields included in selected dialogue template **122/222** (action **583**). For example, the user data for filling the one or more placeholder fields included in selected dialogue template **122/222** may be identified by searching user database **264** on client system **150/250**. That user data may be identified and obtained from user database **264** in action **583** by social agent client application **266**, executed by hardware processor **254** of client system **150/250**.

[0053] Flowchart **580** further includes filling the one or more placeholder fields included in selected dialogue template **122/222** using the user data identified in action **583** to create personalized dialogue **148/248**, using selected dialogue template **122/222**, for responding to user **118** (action **584**). Action **584** may be performed by social agent client application **266**, executed by hardware processor **254** of client system **150/250**, in the manner described above by reference to action **474**.

[0054] Flowchart **580** can continue and conclude with executing personalized dialogue **148/248** (action **585**). Execution of personalized dialogue **148/248** may be performed by social agent client application **266**, executed by hardware processor **254** of client system **150/250**, in a number of different ways, as described below.

[0055] In some implementations, personalized dialogue **148/248** may take the form of language based verbal communication by social agent **116a**. Moreover, in some implementations, output unit **240/340** may include display **158/358**. In those implementations, personalized dialogue **148/248** may be executed by being rendered as text on display **158/358** of client system **150/250**.

[0056] However, in other implementations personalized dialogue **148/248** may include a non-verbal communication by social agent **116a**, either instead of, or in addition to a language based communication. For example, in some implementations, output unit **240/340** may include TTS module **342** and speaker(s) **344**, as well as display **158/358** showing social agent **116a** in the form of a virtual character, such as an avatar or animation, for example. In those implementations, personalized dialogue **148/248** may be executed by being output as speech by the virtual character using speaker(s) **344** of client system **150/250**. In addition, in some implementations, personalized dialogue **148/248** may include a non-verbal vocalization by the virtual character, a facial expression by the virtual character, a gesture by the virtual character, or a physical posture adopted by the virtual character. Alternatively, or in addition, in some implementations, personalized dialogue **148/248** may be accompanied by a haptic effect. In those implementations, executing personalized dialogue **148/248** in action **585** may include activating haptic actuator(s) **347** of client system **150/250**.

[0057] Furthermore, and as shown in FIG. **1**, in some implementations, social agent system **100/200** may include social agent **116b** in the form of a robot or other physical machine capable of simulating expressive behavior and including output unit **240/340**. In those implementations, personalized dialogue **148/248** may be rendered as one or more of speech by social agent **116b**, a

non-verbal vocalization by social agent **116b**, a facial expression by social agent **116b**, a gesture by social agent **116b**, or a physical posture adopted by social agent **116b**, under the control of client system **150/250**.

[0058] For example, in some implementations, hardware processor **254** of client system **150/250** may execute social agent client application **266** to utilize transceiver **252** to establish wireless communication link **115** with social agent **116b**, such as a 4G or 5G wireless communication link, or a wireless communication link implemented using WiFi, Bluetooth, ZigBee, or 60 GHz wireless communications methods, for instance. In those implementations, execution of personalized dialogue **148/248** in action **585** may include wirelessly controlling social agent **116b**, by client system **150/250**, to speak, via speaker(s) **344**, or to move in response to activation of mechanical actuator(s) **346** by client system **150/250**.

[0059] Thus, the present application is directed to systems and methods for personalizing a user interaction with a social agent that address and overcome the deficiencies in the conventional. The inventive concepts disclosed in the present application advantageously enable the determination of naturalistic expressions for use by a social agent in responding to an interaction with a user, while concurrently enabling the user to selectively sequester their personal user data from the system providing the social agent. Moreover, the present solution may advantageously be implemented as automated systems and methods. That is to say, actions **471**, **472**, and **474**, or actions **471**, **472**, **473**, and **474** in FIG. **4**, and actions **581**, **582**, **583**, **584**, and **585** in FIG. **5**, may be performed in automated processes from which human involvement may be omitted. [0060] 1. In some embodiments, a computer-implemented method for processing data comprises determining a privacy level associated with user data that is (i) stored on a client system and (ii) associated with a user; generating, by the client system, input data associated with the user based on (i) the privacy level, and (ii) one or more inputs received via one or more sensors on the client system; transmitting, by the client system, the input data over a network communication link to a computing platform; receiving, from the computing platform in response to the transmitted input data, a dialogue template having at least one placeholder field; determining at least a subset of the user data that is associated with the at least one placeholder field; generating a personalized dialogue by filling the at least one placeholder field using the at least the subset of the user data; and causing the personalized dialogue to be outputted to the user. [0061] 2. The computer-implemented method of clause 1, wherein determining the privacy level comprises receiving a selection of the privacy level from the user. [0062] 3. The computer-implemented method of any of clauses 1-2, wherein generating the input data comprises partially anonymizing the user data based on the privacy level. [0063] 4. The computer-implemented method of any of clauses 1-3, wherein the partially anonymized user data corresponds to personally identifiable information included in the user data. [0064] 5. The computer-implemented method of any of clauses 1-4, wherein causing the personalized dialogue to be outputted to the user comprises rendering text corresponding to the personalized dialogue on a display of the client system. [0065] 6. The computer-implemented method of any of clauses 1-5, wherein causing the personalized dialogue to be outputted to the user comprises outputting the personalized dialogue as speech by a virtual character using one or more speakers of the client system. [0066] 7. The computer-implemented method of any of clauses 1-6, wherein causing the personalized dialogue to be outputted to the user comprises causing at least one of speech, a non-verbal vocalization, a facial expression, a gesture, or a physical posture to be generated by a robot under control of the client system. [0067] 8. The computer-implemented method of any of clauses 1-7, wherein the one or more sensors comprise at least one of a camera, a microphone, a radio-frequency identification sensor, a facial recognition sensor, an automatic speech recognition sensor, or an object recognition sensor. [0068] 9. The computer-implemented method of any of clauses 1-8, wherein the privacy level restricts access to an amount of the user data. [0069] 10. The computer-implemented method of any of clauses 1-9, wherein the input data comprises at least one of speech, text, one or more non-verbal expressions, or one or more manual

inputs to an input device on the client system. [0070] 11. In some embodiments, one or more non-transitory computer-readable media store instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of determining a privacy level associated with user data that is (i) stored on a client system and (ii) associated with a user; generating, by the client system, input data associated with the user based on (i) the privacy level, and (ii) one or more inputs received via one or more sensors on the client system; transmitting, by the client system, the input data over a network communication link to a computing platform; receiving, from the computing platform in response to the transmitted input data, a dialogue template having at least one placeholder field; determining at least a subset of the user data that is associated with the at least one placeholder field; generating a personalized dialogue by filling the at least one placeholder field using the at least the subset of the user data; and outputting the personalized dialogue to the user using one or more output devices associated with the client system. [0071] 12. The one or more non-transitory computer-readable media of clause 11, wherein the instructions further cause the one or more processors to perform the step of receiving a selection of the privacy level from the user prior to generating the input data. [0072] 13. The one or more non-transitory computer-readable media of any of clauses 11-12, wherein generating the input data comprises restricting access by the computing platform to the at least the subset of the user data. [0073] 14. The one or more non-transitory computer-readable media of any of clauses 11-13, wherein generating the input data comprises using a proxy for an identifiable object included in the user data. [0074] 15. The one or more non-transitory computer-readable media of any of clauses 11-14, wherein outputting the personalized dialogue comprises establishing a wireless communication link between the client system and a physical machine; and wirelessly controlling the physical machine to perform at least one of speech or movement corresponding to the personalized dialogue. [0075] 16. The one or more non-transitory computer-readable media of any of clauses 11-15, wherein the one or more output devices comprise at least one of a display, a speaker, or one or more mechanical actuators. [0076] 17. The one or more non-transitory computer-readable media of any of clauses 11-16, wherein the input data comprises at least one of audio, video, motion capture data, a gesture, a physical posture, a non-verbal vocalization, or a facial expression. [0077] 18. In some embodiments, a system comprises one or more memories that store instructions, and one or more processors that are coupled to the one or more memories and, when executing the instructions, are configured to perform the steps of receiving, over a network communication link from a client system, input data associated with a user of the client system, wherein the input data is generated by the client system based on (i) user data stored on the client system and (ii) a privacy level associated with the user data; selecting, based on the input data, a dialogue template having at least one placeholder field; and transmitting the dialogue template to the client system, wherein the client system generates and outputs a personalized dialogue to the user by filling the at least one placeholder field using at least a subset of the user data to which access is selectively restricted based on the privacy level. [0078] 19. The system of clause 18, wherein selecting the dialogue template comprises determining, using a machine learning predictive model based on the input data, a goal of the user; and selecting the dialogue template based on the goal. [0079] 20. The system of any of clauses 18-19, wherein the goal is determined based on at least one of a verbal expression included in the input data, a non-verbal expression included in the input data, or a subject matter of an interaction described by the input data. [0080] From the above description it is manifest that various techniques can be used for implementing the concepts described in the present application without departing from the scope of those concepts. Moreover, while the concepts have been described with specific reference to certain implementations, a person of ordinary skill in the art would recognize that changes can be made in form and detail without departing from the scope of those concepts. As such, the described implementations are to be considered in all respects as illustrative and not restrictive. It should also be understood that the present application is not limited to the particular implementations described

herein, but many rearrangements, modifications, and substitutions are possible without departing from the scope of the present disclosure.

Claims

- 1.** A computer-implemented method for processing data, the method comprising: determining a privacy level associated with user data that is (i) stored on a client system and (ii) associated with a user; generating, by the client system, input data associated with the user based on (i) the privacy level, and (ii) one or more inputs received via one or more sensors on the client system; transmitting, by the client system, the input data over a network communication link to a computing platform; receiving, from the computing platform in response to the transmitted input data, a dialogue template having at least one placeholder field; determining at least a subset of the user data that is associated with the at least one placeholder field; generating a personalized dialogue by filling the at least one placeholder field using the at least the subset of the user data; and causing the personalized dialogue to be outputted to the user.
- 2.** The computer-implemented method of claim 1, wherein determining the privacy level comprises receiving a selection of the privacy level from the user.
- 3.** The computer-implemented method of claim 1, wherein generating the input data comprises partially anonymizing the user data based on the privacy level.
- 4.** The computer-implemented method of claim 3, wherein the partially anonymized user data corresponds to personally identifiable information included in the user data.
- 5.** The computer-implemented method of claim 1, wherein causing the personalized dialogue to be outputted to the user comprises rendering text corresponding to the personalized dialogue on a display of the client system.
- 6.** The computer-implemented method of claim 1, wherein causing the personalized dialogue to be outputted to the user comprises outputting the personalized dialogue as speech by a virtual character using one or more speakers of the client system.
- 7.** The computer-implemented method of claim 1, wherein causing the personalized dialogue to be outputted to the user comprises causing at least one of speech, a non-verbal vocalization, a facial expression, a gesture, or a physical posture to be generated by a robot under control of the client system.
- 8.** The computer-implemented method of claim 1, wherein the one or more sensors comprise at least one of a camera, a microphone, a radio-frequency identification sensor, a facial recognition sensor, an automatic speech recognition sensor, or an object recognition sensor.
- 9.** The computer-implemented method of claim 1, wherein the privacy level restricts access to an amount of the user data.
- 10.** The computer-implemented method of claim 1, wherein the input data comprises at least one of speech, text, one or more non-verbal expressions, or one or more manual inputs to an input device on the client system.
- 11.** One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of: determining a privacy level associated with user data that is (i) stored on a client system and (ii) associated with a user; generating, by the client system, input data associated with the user based on (i) the privacy level, and (ii) one or more inputs received via one or more sensors on the client system; transmitting, by the client system, the input data over a network communication link to a computing platform; receiving, from the computing platform in response to the transmitted input data, a dialogue template having at least one placeholder field; determining at least a subset of the user data that is associated with the at least one placeholder field; generating a personalized dialogue by filling the at least one placeholder field using the at least the subset of the user data; and outputting the personalized dialogue to the user using one or more output devices associated

with the client system.

12. The one or more non-transitory computer-readable media of claim 11, wherein the instructions further cause the one or more processors to perform the step of receiving a selection of the privacy level from the user prior to generating the input data.

13. The one or more non-transitory computer-readable media of claim 11, wherein generating the input data comprises restricting access by the computing platform to the at least the subset of the user data.

14. The one or more non-transitory computer-readable media of claim 11, wherein generating the input data comprises using a proxy for an identifiable object included in the user data.

15. The one or more non-transitory computer-readable media of claim 11, wherein outputting the personalized dialogue comprises: establishing a wireless communication link between the client system and a physical machine; and wirelessly controlling the physical machine to perform at least one of speech or movement corresponding to the personalized dialogue.

16. The one or more non-transitory computer-readable media of claim 11, wherein the one or more output devices comprise at least one of a display, a speaker, or one or more mechanical actuators.

17. The one or more non-transitory computer-readable media of claim 11, wherein the input data comprises at least one of audio, video, motion capture data, a gesture, a physical posture, a non-verbal vocalization, or a facial expression.

18. A system, comprising: one or more memories that store instructions, and one or more processors that are coupled to the one or more memories and, when executing the instructions, are configured to perform the steps of: receiving, over a network communication link from a client system, input data associated with a user of the client system, wherein the input data is generated by the client system based on (i) user data stored on the client system and (ii) a privacy level associated with the user data; selecting, based on the input data, a dialogue template having at least one placeholder field; and transmitting the dialogue template to the client system, wherein the client system generates and outputs a personalized dialogue to the user by filling the at least one placeholder field using at least a subset of the user data to which access is selectively restricted based on the privacy level.

19. The system of claim 18, wherein selecting the dialogue template comprises: determining, using a machine learning predictive model based on the input data, a goal of the user; and selecting the dialogue template based on the goal.

20. The system of claim 19, wherein the goal is determined based on at least one of a verbal expression included in the input data, a non-verbal expression included in the input data, or a subject matter of an interaction described by the input data.
